

Plane Geometry

ANSWER KEY



Angle relationships

- Two angles are supplementary, and one is 3 times the other. Find both angles.
Let the angles be x and $3x$: $x + 3x = 180$, so $4x = 180$, giving $x = 45$. The angles are 45° and 135° .
- Two parallel lines are cut by a transversal, creating an angle of 70° . Find the measure of its alternate interior angle and its corresponding angle.
Alternate interior angles are equal: 70° . Corresponding angles are also equal: 70° .

Triangle angle sum and exterior angles

- A triangle has angles measuring x , $2x$, and $3x$. Find each angle.
 $x + 2x + 3x = 180$, so $6x = 180$, giving $x = 30$. Angles: 30° , 60° , 90° .
- A triangle's exterior angle at one vertex measures 115° . If one of the remote interior angles is 50° , find the other remote interior angle.
By the exterior angle theorem, the exterior angle equals the sum of the two remote interior angles:
 $115 = 50 + x$, so $x = 65^\circ$.

Area and perimeter of polygons

- Find the area of a triangle with base 14 and height 9.
 $\text{Area} = \frac{1}{2}(14)(9) = 63$.
- Find the area of a parallelogram with base 12 and height 5.
 $\text{Area} = \text{base} \times \text{height} = 12 \times 5 = 60$.
- Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.
 $A = \frac{3\sqrt{3}}{2}(6)^2 = \frac{3\sqrt{3}}{2}(36) = 54\sqrt{3}$.

The Pythagorean theorem and special right triangles

- A right triangle has legs 9 and 12. Find the hypotenuse.
 $c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.
- A 45-45-90 triangle has a leg of length 8. Find the length of the hypotenuse.
In a 45-45-90 triangle, the hypotenuse equals $\text{leg} \times \sqrt{2}$: $8\sqrt{2}$.

- 10 A 30-60-90 triangle has a hypotenuse of 20. Find the lengths of both legs.
In a 30-60-90 triangle, the shorter leg is $\frac{\text{hypotenuse}}{2}$ and the longer leg is $\frac{\text{hypotenuse}}{2} \times \sqrt{3}$: shorter leg = 10, longer leg = $10\sqrt{3}$.
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Circles: area, circumference, arcs, and sectors

- 11 Find the area and circumference of a circle with diameter 14, in terms of π .
Radius = 7. Area = $\pi r^2 = 49\pi$. Circumference = $\pi d = 14\pi$.
- 12 Find the arc length of a 120° sector in a circle with radius 9, in terms of π .
Arc length = $\frac{120}{360} \times 2\pi(9) = \frac{1}{3} \times 18\pi = 6\pi$.
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Similar and congruent figures

- 13 Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.
Areas of similar figures scale by the SQUARE of the linear scale factor: $\left(\frac{5}{3}\right)^2 = \frac{25}{9}$. Larger triangle's area: $18 \times \frac{25}{9} = 50$.
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Volume and surface area of 3D solids

- 14 Find the volume of a rectangular prism with dimensions 5, 8, and 10.
Volume = $5 \times 8 \times 10 = 400$.
- 15 Find the volume of a cone with radius 6 and height 10, in terms of π .
Volume = $\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$.
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Multi-step geometry problems

- 16 This question has two parts. A square garden has area 144 square feet. (a) Find the side length and perimeter of the garden. (b) A circular fountain with radius 3 feet is placed inside the garden. Find the remaining garden area (not covered by the fountain), in terms of π .
(a) Side length: $\sqrt{144} = 12$ feet. Perimeter: $4(12) = 48$ feet. (b) Fountain area: $\pi(3)^2 = 9\pi$. Remaining area: $144 - 9\pi$ square feet.
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Trapezoids and composite polygon areas

- 17 Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

$$\text{Area} = \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = 66.$$

- 18 An L-shaped figure is formed by a 10-by-8 rectangle with a 4-by-3 rectangle removed from one corner. Find the remaining area.

Large rectangle area: $10 \times 8 = 80$. Removed rectangle area: $4 \times 3 = 12$. Remaining area: $80 - 12 = 68$.

Chords, tangents, and inscribed angles in circles

- 19 An inscribed angle subtends an arc of 100° . Find the measure of the inscribed angle.

An inscribed angle is always half the arc it subtends: $\frac{100^\circ}{2} = 50^\circ$.

- 20 A tangent line touches a circle at point P . Explain the relationship between the tangent line and the radius drawn to point P .

A tangent line is always perpendicular to the radius drawn to the point of tangency — this is a fundamental property relating tangent lines and circles.

Congruence and triangle proofs

- 21 State the triangle congruence criterion (SSS, SAS, ASA, AAS) that applies when two triangles share two equal angles and the included side.

ASA (Angle-Side-Angle) applies, since two angles and the side directly between them are all given as congruent.

A multi-part composite geometry problem

- 22 This question has two parts. A semicircular window has a diameter of 12 feet and sits atop a rectangular window of width 12 feet and height 5 feet. (a) Find the area of the semicircular portion, in terms of π . (b) Find the total area of the combined window shape.

(a) Radius = 6. Semicircle area = $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(36) = 18\pi$. (b) Rectangle area: $12 \times 5 = 60$. Total area: $60 + 18\pi$ square feet.